DG-TDM: DUAL-GRANULARITY TEMPORAL DYNAMICS MODELING WITH SPATIAL PRIOR GUIDANCE FOR FINE-GRAINED ACTION RECOGNITION

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ABSTRACT

Fine-grained action recognition requires distinguishing actions that look nearly identical in single frames and differ mainly in how the body moves. Two difficulties follow from this setting. Since all classes share a nearly identical background, the action region must be located rather than inferred from scene cues; and since the discriminating signal is motion, it must be captured at two time scales, the short-term displacement of body parts over a few frames and the long-term evolution of the whole sequence. We propose DG-TDM, a three-branch framework that assigns these two requirements to separate branches. A frozen CLIP encoder serves as a spatial expert, and a spatio-temporal guidance (STG) branch injects its stable priors into a lightweight 2D temporal encoder through bilateral window cross-attention and adaptive gating, so that the encoder concentrates on the action region. The core of the framework is a dual-granularity temporal dynamics module that adds temporal modeling to the 2D backbone at the two time scales: the MoDeA branch predicts dense cross-frame sampling offsets to follow short-term body-part displacement, and the TSA branch maintains a long-term state memory to model the sequence-level evolution. We evaluate the framework on Diving48 and Something-Something V2 and use ablations to isolate the contribution of each component.

1 INTRODUCTION

Fine-grained action recognition distinguishes actions that look nearly identical in single frames and differ mainly in how the body moves. Recent benchmarks drive the task toward this setting: Diving48 Li et al. (2018) separates dives by their rotation and timing, and Something-Something V2 Goyal et al. (2017) separates everyday object manipulations by the direction and manner of the motion, such as pushing something from left to right versus from right to left. In both datasets, the classes are not separated by the scene or by whole-body appearance but by how a small set of body parts moves over time. On Diving48, for example, a 720° rotation and a 900° rotation of the same dive are two classes, yet every frame of the two videos shows the same diver in the same poses against the same pool background; only across frames does the rotation itself become visible.

This setting raises two problems. The first is spatial: since all classes share the same background, the background carries no label-relevant information and only distracts the model, so recognition must come from the region where the body moves. The model thus has to know where to look before it can read the motion. The second is temporal, and it spans two scales. On the short scale, a class is often decided by a small local displacement over a few frames, such as the path of a hand or the change in the angle of a limb. The position that carries this detail moves from frame to frame, so a fixed sampling grid loses it. On the long scale, the full meaning of an action is determined by how the entire sequence evolves, so the model must maintain a state over time rather than treat each frame in isolation. Both scales matter, and dropping either one loses evidence that separates otherwise similar classes.

Current video models only partially meet these needs. 3D CNNs Carreira and Zisserman (2017); Feichtenhofer et al. (2019) and video transformers trained on large corpora Bertasius et al. (2021); Liu et al. (2022); Tong et al. (2022); Wang et al. (2024a) capture temporal structure, but they compute

features at every position of every frame, spending most of their capacity on background regions that carry no signal; moreover, their sampling and attention are either too rigid to follow a moving body part or too coarse to resolve a small displacement. Deformable video transformers Wang and Torresani (2022) learn where to attend and perform well on fine-grained benchmarks, but they select whole patches and therefore miss the sub-window displacement that separates fine-grained classes. CLIP Radford et al. (2021) is strong precisely where the first problem lies: pre-trained on massive image–text data, it judges reliably where people and action-relevant content appear in a frame. Yet when CLIP is adapted to video, either through frozen adapters Lin et al. (2022) or end-to-end fine-tuning Rasheed et al. (2023), the prior is absorbed into the encoder itself and is not used to tell a separate recognition branch where the action is. State-space models Gu and Dao (2023); Park et al. (2024) aggregate long sequences efficiently, but the spatial side remains open.

These observations suggest a division of labor. A frozen spatial expert answers where the action is, and a trainable temporal encoder learns how it moves. Because the expert is frozen, its positional judgment stays constant throughout training and cannot overfit the background statistics of a small fine-grained dataset; because the encoder is trained end-to-end, it can shape its features around the motion. The two sides are complementary rather than redundant.

Following this idea, we propose DG-TDM, a three-branch framework. A frozen CLIP encoder serves as the spatial expert, and a spatio-temporal guidance (STG) branch transfers its priors to the temporal encoder through bilateral window cross-attention and adaptive gating, so that the encoder focuses on the action region. The temporal encoder is built on a 2D ResNet backbone, which has no temporal modeling of its own. We therefore insert, at every stage, a dual-granularity temporal dynamics module (DG-TDM) that adds temporal modeling on the 2D feature stream without inflating the backbone into a 3D network. This module gives the overall framework its name, so we use the same acronym for both throughout the paper. DG-TDM works at the two time scales of the motion: its MoDeA branch predicts dense cross-frame sampling offsets to follow short-term body-part displacement, and its TSA branch carries a long-term state memory over the sequence to model how the action develops.

In summary, our main contributions are:

- We propose DG-TDM, a three-branch framework for fine-grained action recognition that assigns localization to a frozen CLIP spatial expert and motion modeling to a trainable temporal encoder.
- We design a spatio-temporal guidance branch that injects the frozen priors into the temporal encoder through bilateral window cross-attention and adaptive gating, so that the encoder focuses on the action region.
- We develop a dual-granularity temporal dynamics module for the 2D backbone, in which MoDeA tracks short-term body-part displacement through dense deformable sampling and TSA models long-term evolution through a bidirectional state memory.

2 Related Work

Video modeling. Early approaches adapt image architectures to video by fusing frames at the input or inflating convolutions to 3D Lin et al. (2018); Carreira and Zisserman (2017). Video transformers later replaced fixed kernels with self-attention, spanning divided space-time attention in TimeSformer Bertasius et al. (2021), pooled multiscale attention in MViTv2 Li et al. (2022a), unified convolution–attention designs in UniFormer Li et al. (2022b), spatio-temporal focal modulation in Video-FocalNets Wasim et al. (2023), and hierarchical windowed attention in Video Swin Liu et al. (2022). On the training side, masked reconstruction Tong et al. (2022), BERT-style video pretraining Wang et al. (2022), and object-centric attention Girdhar et al. (2022) enrich the learned representations, and general video foundation models Wang et al. (2024a) scale this further. These models are powerful but compute features at every position of every frame, which is wasteful when only a small region moves, as in fine-grained action. Deformable approaches instead learn where to sample: the Deformable Video Transformer Wang and Torresani (2022) dynamically predicts a small subset of video patches to attend, achieving efficient temporal aggregation. Our MoDeA branch follows this direction but predicts a dense 3D offset field and performs continuous trilinear sampling, capturing sub-window motion that patch-level selection cannot represent.

Adapters and parameter-efficient video models. Video adapters insert lightweight modules into a frozen encoder to adapt it to a downstream task. ST-Adapter Pan et al. (2022) pioneered this direction for video, inserting space–time adapters between the blocks of a frozen image backbone and showing that a small trainable footprint suffices for recognition. This idea underlies EVL Lin et al. (2022), which attaches temporal adapters to a fully frozen CLIP encoder, and AIM Yang et al. (2023), which adapts an image backbone to video with spatial–temporal modules and reaches competitive accuracy at a small training cost. These designs keep the pretrained weights frozen and add a small trainable footprint, which is valuable when fine-grained training data is limited. We follow the same adapter-style residual design and extend it in two respects: a dual-granularity temporal module that pairs local deformable motion tracking with a long-term memory of the temporal response, and a guidance branch that composes the frozen prior with a separate trainable video backbone.

State-space models and long-term temporal memory. Mamba Gu and Dao (2023) models long sequences with a learned recurrent state at linear cost, and video variants extend it along the temporal axis: VideoMamba Li et al. (2024) scans spatio-temporal tokens for efficient video understanding, and a concurrent VideoMamba Park et al. (2024) introduces a bidirectional spatio-temporal selective scan. Our TSA branch shares the same long-term memory intuition, carrying a state across the sequence to model the overall response of an action, but instantiates it in a lightweight, channel-wise form: a bidirectional linear recurrence with a per-channel decay, used to produce a temporal channel gate rather than to replace the whole architecture. This gives the model a cheap way to remember how each channel responds over the whole action, complementing the local motion tracking of MoDeA.

Frozen vision-language priors. Contrastive image–text models such as CLIP Radford et al. (2021) acquire strong knowledge of where humans and action-relevant content appear, learned from massive pre-training data. ViFi-CLIP Rasheed et al. (2023) shows that fine-tuning CLIP on videos transfers this knowledge effectively when downstream data is scarce, while EVL Lin et al. (2022) demonstrates the frozen-encoder route. We follow the frozen route, since a frozen expert guarantees positional judgments that never drift with the downstream task, and we differ in how the prior is used: rather than absorbing the prior into the encoder, we route it through a guidance branch that learns where and how much to inject. The bilateral cross-attention lets the prior and the lower branch refine each other, and the adaptive gate controls the amount of injection per position, which is better suited to fine-grained tasks where indiscriminate fusion can dilute the discriminative signal.

Fine-grained action recognition. Fine-grained action recognition targets classes that differ in subtle body movement rather than in scene Li et al. (2018). The difficulty lies both in localizing the small moving region and in representing the fine movement itself. Recent methods learn specialized spatio-temporal experts for this task: DSTS Li et al. (2022c) lets each stage specialize in the discriminative spatio-temporal scales of a class, and TDN Wang et al. (2021) models temporal difference at multiple scales. These methods focus on temporal modeling but treat localization as an implicit side effect. Our framework separates the two concerns explicitly: the frozen prior handles localization through the guidance branch, while the dual-granularity module handles representation through deformable motion and state aggregation.

3 METHOD

We build the framework around two observations. Because the background is shared across classes and carries no label-relevant information, the model needs a reliable way to locate the action; because the classes differ in how the body moves, the model must also capture the motion at both short and long time scales. The framework, shown in Figure 2, assigns these needs to three cooperating branches: a frozen CLIP encoder that locates the action, a spatio-temporal guidance (STG) block that transfers this knowledge to the recognition branch, and a temporal action encoder, built on a 2D ResNet backbone, that models the motion through its dual-granularity temporal dynamics (DG-TDM) module. Figure 1 zooms in on the two core blocks, STG and DG-TDM.

The data flow is as follows. Given T sampled frames, the frozen CLIP encoder extracts a patch map from each frame, and a lightweight alignment module adapts it to the resolution of every backbone stage. The lower branch is the recognition path: the 2D backbone extracts features stage by stage, and after each of the four stages the STG block injects the aligned prior into the features through bilateral window cross-attention, while the DG-TDM module, inserted after the last bottleneck of

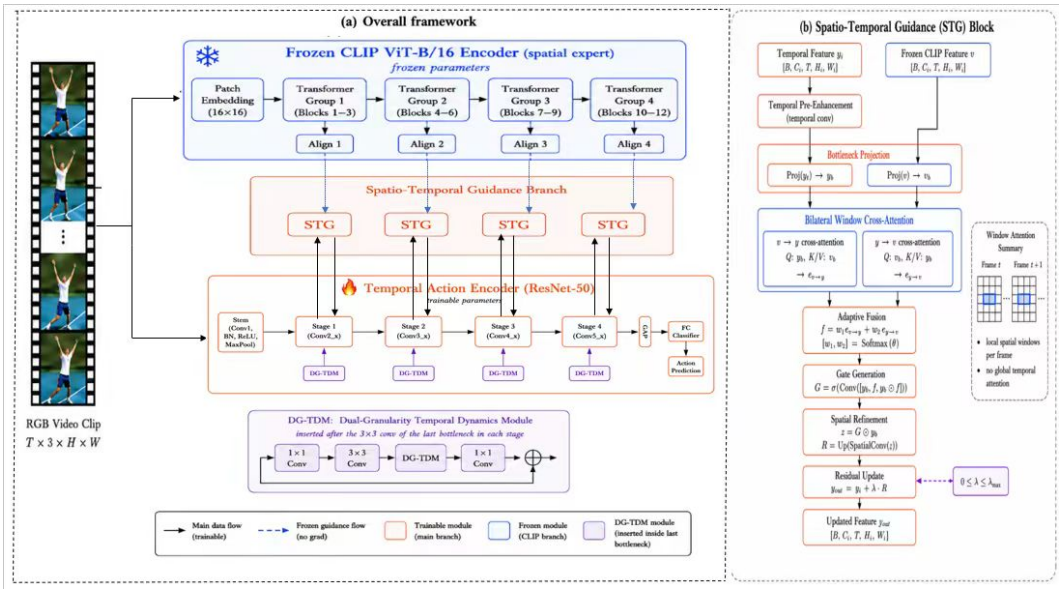


Figure 1: Left: the dual-granularity temporal dynamics module (DG-TDM), composed of the MoDeA branch (motion deformation attention) and the TSA branch (temporal state aggregation). Right: the STG block with bilateral window cross-attention, adaptive fusion, and gated refinement.

the stage, adds temporal modeling on the 2D feature stream. The final stage feature is pooled and classified. The three branches divide the work clearly: the frozen expert answers where the action is, the guidance decides how much of this knowledge each position receives, and the temporal encoder models how the action moves. The rest of this section details each part.

Notation. Let a video be sampled into T frames and processed by a 2D backbone with L stages. At stage ℓ, the lower branch produces a feature map $Y_\ell \in \mathbb{R}^{B\times C_\ell \times T\times H_\ell \times W_\ell}$, where B is the batch size and (H_ℓ, W_ℓ) is the spatial resolution. The frozen CLIP encoder produces a patch map $V \in \mathbb{R}^{B\times 768\times T\times 14\times 14}$ that is aligned to each stage resolution by a lightweight projection $\mathrm{AuxAlign}_\ell(\cdot)$, yielding $V_\ell \in \mathbb{R}^{B\times C_\ell \times T\times H_\ell \times W_\ell}$. We use $\odot$ for element-wise multiplication and $\oplus$ for residual addition, DWConv_T for a depth-wise convolution along time, and GAP_{hw} for global average pooling over the spatial dimensions.

3.1 Spatial prior and spatio-temporal guidance

Spatial prior. The upper branch is a frozen CLIP ViT-B/16 encoder Radford et al. (2021). What we use is not object semantics but the spatial-position discriminability acquired during pre-training: the encoder learns where humans, body parts, and action-relevant objects tend to appear, and can judge, at each location, whether the content belongs to the action or the background. The encoder operates frame-wise: each frame passes through the ViT independently, the class token is discarded, and the patch tokens form a 768-dimensional patch map V of resolution 14×14. Since the encoder is frozen, this positional knowledge enters as a stable prior that cannot be distorted by the limited fine-grained training set. To make the prior available at every stage, a lightweight alignment module $\mathrm{AuxAlign}_\ell$ interpolates V to (T, H_ℓ, W_ℓ) and applies a $1 \times 1 \times 1$ convolution, batch normalization, and ReLU; the aligned prior V_ℓ feeds the guidance branch below, and these alignment modules are the only trainable parameters attached to the frozen encoder.

Spatio-temporal guidance. The STG block is inserted after each temporal stage. Its purpose is to make the lower branch attend to the action regions that the frozen encoder identifies, so that the lower branch is not distracted by the background. We use bilateral window cross-attention: information flows both from the prior into the lower branch and from the lower branch into the prior, so that the fusion is a mutual refinement rather than a one-way injection (Figure 1, right).

Given the lower-branch feature Y_ℓ and the aligned prior V_ℓ, the STG block first applies a light temporal convolution to Y_ℓ to pre-enhance temporal cues, then projects both inputs into a shared bottleneck dimension C_ℓ/r through separate $1 \times 1 \times 1$ convolutions:

$$Y_b = \mathrm{Conv}^1_{\mathrm{down}}(Y_\ell), \qquad V_b = \mathrm{Conv}^1_{\mathrm{down}}(V_\ell). \tag{1}$$

The projection to a lower dimension reduces the cost of the subsequent attention and forces the two inputs to share a compact representation before interaction.

We then compute bilateral window cross-attention within each local spatial window. The window operation is confined to $w \times w$ neighborhoods in every frame and never mixes across time, which keeps the cost linear in the spatial resolution and preserves per-frame spatial structure: the feature is reshaped into non-overlapping windows, the attention is applied inside each window, and the result is restored to the original layout.

The first direction, $V \to Y$, treats Y_b as queries and V_b as keys and values, letting the lower branch query the frozen prior:

$$e_{V\to Y} = \mathrm{Softmax}\Big(\frac{Q_Y K_V^\top}{\sqrt{d}}\Big)V_V, \qquad Q_Y = Y_b W_Q,\ K_V = V_b W_K,\ V_V = V_b W_V, \tag{2}$$

where W_Q, W_K, W_V are the query, key, and value projections and d is the head dimension. The second direction, $Y \to V$, reverses the roles and lets the prior query the lower branch, so that the guidance reflects what the lower branch has already learned:

$$e_{Y\to V} = \mathrm{Softmax}\Big(\frac{Q_V K_Y^\top}{\sqrt{d}}\Big)V_Y, \qquad Q_V = V_b W_Q,\ K_Y = Y_b W_K,\ V_Y = Y_b W_V. \tag{3}$$

Multi-head attention is applied in both directions, with the query, key, and value split across heads so that each head can attend to a different aspect of the interaction.

The two directional outputs are combined by a learned softmax-weighted fusion to adaptively balance the contribution of each direction. Rather than a fixed average, we let the network learn the relative weight of the two directions, which allows the guidance to be dominated by the prior when the lower branch is uncertain and by the lower branch when it has learned reliable features:

$$[w_1, w_2] = \mathrm{Softmax}(\beta), \qquad f = w_1\, e_{V\to Y} + w_2\, e_{Y\to V}, \tag{4}$$

where $\beta \in \mathbb{R}^2$ is initialized so that $w_1 = w_2 = 0.5$. A gating network then decides, per spatial position, how much of the fused prior should be applied to the lower branch. The gate takes the concatenation of Y_b, the fused feature f, and their element-wise product, and outputs a value in $(0, 1)$:

$$G = \sigma\big(\mathrm{Conv}\big([\,Y_b,\ f,\ Y_b \odot f\,]\big)\big). \tag{5}$$

Including the product term $Y_b \odot f$ gives the gate access to the interaction between the two inputs, enabling it to suppress positions where they disagree. The gate is initialized with a negative bias, so that the guidance is weak early in training and the lower branch can stabilize first. The gated feature is refined by a spatial convolution and up-projected back to the stage channel count, then added to the lower-branch feature with a bounded residual scale $\lambda \in [0, \lambda_{\max}]$:

$$\tilde{Y}_\ell = Y_\ell + \lambda \cdot \mathrm{Up}\big(G \odot Y_b\big). \tag{6}$$

The scale λ is a learnable scalar parameterized through a logit so that it stays within $[0, \lambda_{\max}]$; it is initialized small to preserve the identity at the start of training.

3.2 Dual-granularity temporal dynamics module

The lower branch is a temporal action encoder built on a 2D ResNet backbone. A 2D backbone computes the feature of each frame independently and has no temporal modeling of its own; rather than inflating it into a 3D network, we insert the dual-granularity temporal dynamics module (DG-TDM) (Figure 1, left) after the 3×3 convolution of the last bottleneck in each of the four stages. DG-TDM is a pseudo-temporal module: it does not process frames jointly like a 3D convolution, but injects temporal dynamics into the 2D features through deformable cross-frame sampling and a temporal state scan, leaving the backbone structure and its pretrained weights intact. The module

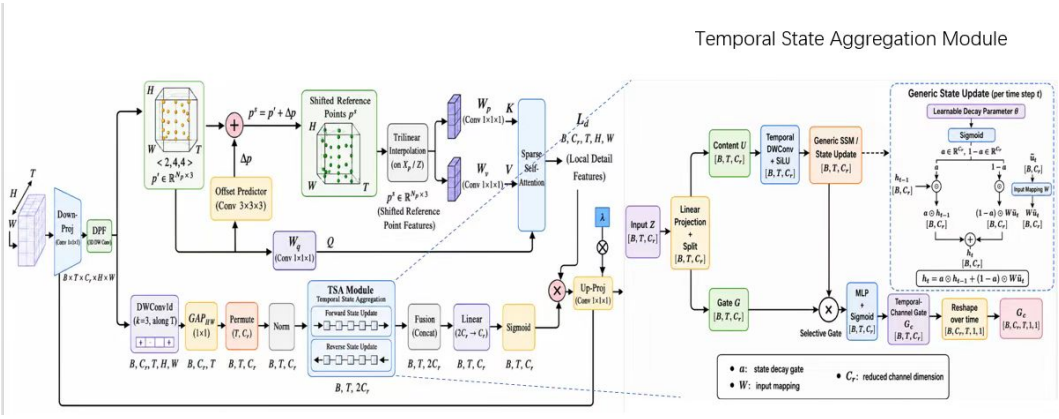


Figure 2: Overall framework of DG-TDM. A frozen CLIP encoder extracts spatial priors from each frame; the spatial-temporal guidance (STG) branch injects these priors into a temporal action encoder through bilateral window cross-attention and adaptive gating. Inside each temporal stage, the dual-granularity temporal dynamics module (DG-TDM) refines the feature with a motion deformation attention (MoDeA) branch and a temporal state aggregation (TSA) branch.

models the motion at the two time scales identified in the introduction. MoDeA handles the short-term part: it predicts where to sample across frames to follow the displacement of a body part over successive frames. TSA handles the long-term part: it carries a state over the whole sequence and models how the overall temporal response of the action develops. Together they let the lower branch represent the fine movements that distinguish fine-grained actions.

3.2.1 Adapter-style formulation

Following an adapter-style residual design, the DG-TDM block first projects the stage feature into a compact bottleneck dimension C_ℓ/r, then applies the two motion branches, and finally projects back and adds a residual. To preserve the identity at initialization, the output projection is zero-initialized and a learnable scalar scale is applied. The structure is

$$X_d = \mathrm{Down}(X), \tag{7}$$
$$X_p = X_d + \mathrm{DPE}(X_d), \tag{8}$$
$$L_d = X_p + \mathrm{MoDeA}\big(\mathrm{LN}(X_p)\big), \tag{9}$$
$$\tilde{G}_c = \mathrm{TSA}\big(\mathrm{GAP}_{hw} \circ \mathrm{DWConv}_T(X_p)\big), \tag{10}$$
$$\tilde{X} = X + s \cdot \mathrm{Up}\big(\tilde{G}_c \odot L_d\big), \tag{11}$$

where Down and Up are $1 \times 1 \times 1$ convolutions, DPE is a depth-wise $3 \times 3 \times 3$ convolution that injects dynamic position information, LN is layer normalization applied per spatial–temporal position, DWConv_T is a depth-wise convolution along time, GAP_{hw} is global average pooling over the spatial dimensions, and s is the learnable scale. We describe MoDeA and TSA next.

The bottleneck design keeps the DG-TDM cheap: the two motion branches operate in C_ℓ/r channels rather than C_ℓ, and the output projection is zero-initialized so that the block behaves as an identity at the start of training. This is important for fine-grained tasks because it lets the pretrained backbone be reused directly and adds only a small trainable footprint; it also stabilizes early optimization, since the block initially passes the stage feature through unchanged and only gradually learns to add motion-aware refinements. The two branches read the same projected input X_p but in different ways: MoDeA operates on the full spatio-temporal tensor and refines it locally, while TSA pools the spatial dimensions first and returns a compact per-channel gate. The gate is applied to the MoDeA output L_d before the up-projection, so the two branches interact as a product: the local motion details are kept only where the temporal memory indicates that the channel matters for the action.

3.2.2 MOTION DEFORMATION ATTENTION (MODEA)

Actions are, by definition, movements of body parts relative to one another: a hand travels from the upper-left corner of the frame to somewhere else, and the trajectory of that movement is what distinguishes one action from another. A key consequence is that the position of the action content is *not* fixed across frames: the location that carries the discriminative detail at time t holds different, irrelevant content by time $t+1$. A feature sampled on a fixed grid therefore loses the motion, because it keeps sampling the same coordinates while the content has moved away. MoDeA addresses this by predicting, at each spatio-temporal position, a deformation offset that tracks where the relevant content has moved, and then attending over the displaced sampling positions. This follows the line of deformable video transformers Wang and Torresani (2022), which select the patches to attend based on motion, but differs in that MoDeA predicts a dense 3D offset field and performs continuous trilinear sampling, giving a finer representation of local motion than patch-level selection.

Formally, let $Z = \mathrm{LN}(X_p)$ denote the layer-normalized input of shape $[B, C_r, T, H, W]$, where $C_r = C_\ell / r$. The branch first projects the query from Z:

$$Q = ZW_Q, \tag{12}$$

with W_Q a $1 \times 1 \times 1$ convolution. The query is then split into G channel groups, and each group produces a deformation field through a depth-wise 3D convolution followed by layer normalization, GELU, and a point-wise convolution that outputs three channels:

$$\widehat{\Delta p} = \mathrm{Conv}_{1\times1}\Big(\mathrm{GELU}\big(\mathrm{LN}\big(\mathrm{DWConv}_{3d}(Q)\big)\big)\Big), \tag{13}$$

where $\widehat{\Delta p} \in \mathbb{R}^{B\cdot G\times 3\times T_p\times H_p\times W_p}$ encodes offsets $(\Delta z, \Delta y, \Delta x)$ at the down-sampled lattice (T_p, H_p, W_p). The raw offsets are passed through tanh and scaled so that the sampling stays stable:

$$\Delta p = \tanh\big(\widehat{\Delta p}\big) \odot \rho, \qquad \rho_d = \min\Big(1,\ \frac{f}{S_d}\Big), \tag{14}$$

where S_d is the lattice size along dimension d and f is a scaling factor. This bounds the displacement to a fraction of the lattice, preventing the sampling from drifting too far from the reference.

A reference grid p is defined on the down-sampled spatio-temporal lattice, normalized to $[-1, 1]$, and expanded over the batch and groups:

$$p_{ijk} = \Big(\frac{2i}{T_p} - 1,\ \frac{2j}{H_p} - 1,\ \frac{2k}{W_p} - 1\Big). \tag{15}$$

The actual sampling positions are $p' = p + \Delta p$. The input feature is sampled at p' with trilinear interpolation:

$$X_s = \mathrm{GridSample}\big(Z,\ p'\big), \tag{16}$$

which yields the sampled feature X_s at the deformed positions. Because sampling is continuous, X_s captures motion that falls between grid cells.

The sampled features are projected into keys and values, and a multi-head attention is computed between the full query positions and the sampled positions:

$$\mathrm{MoDeA}(Z) = \mathrm{Softmax}\Big(\frac{QK^\top}{\sqrt{d}}\Big)V_a, \tag{17}$$

with Q from Z, and K and V_a from X_s. The attention aggregates the sampled positions for every query, letting each query collect motion-deformed context.

Because each group has its own displacement field, different groups can specialize to different motion directions or magnitudes. One group may track a fast-moving hand while another tracks a slowly translating torso, and the attention over the group-specific sampled positions integrates the context that each group needs. Because the offsets are predicted from the data itself, no supervision or optical flow is required; the deformation emerges purely from the recognition objective. This branch is what gives the lower branch the ability to represent motion change: it turns the fixed-grid feature into one that actively follows where the informative content has moved between frames.

3.2.3 TEMPORAL STATE AGGREGATION (TSA)

The long scale calls for a different mechanism. Two classes may share the same short-term displacement pattern and differ only in how their responses accumulate over the entire sequence, and this difference is exactly what a local sample cannot reveal. TSA therefore maintains a state over time and aggregates evidence across all frames rather than within a local neighborhood. It is built as a lightweight selective state-space module (Figure 1, left, bottom-right). In brief, it splits the channel-wise temporal signal into a content stream U, which the state update accumulates into memory, and a selection gate G, which decides what is kept; the content stream passes through a bidirectional state update whose learnable decay controls the memory horizon, the state output is multiplied by the gate, and the result becomes a temporal channel gate that reweights the lower-branch feature.

Linear projection and split. Given the input $Z \in \mathbb{R}^{B\times T\times C_r}$ (after spatial pooling and a normalization), TSA first applies a linear projection and splits the output along the channel dimension into two equal streams:

$$[\,U,\ G\,] = \mathrm{Split}\big(\mathrm{Linear}(Z)\big), \qquad U, G \in \mathbb{R}^{B\times T\times C_r}. \tag{18}$$

The two streams play complementary roles. The content stream U is the signal to be memorized: it is what the state update accumulates over time, so after the recurrence, U determines what the long-term memory contains. The gate stream G is the selector: it decides, at each time step and for each channel, how much of the memorized content is kept. Splitting one projection into both lets them specialize: U learns an input that aggregates well over time, while G learns an input that marks the time steps worth keeping.

Context enhancement. The content stream U is first passed through a depth-wise temporal convolution and a SiLU activation to produce the context $\tilde{U}$:

$$\tilde{U} = \mathrm{SiLU}\big(\mathrm{DWConv}_T(U)\big), \qquad \tilde{U} \in \mathbb{R}^{B\times T\times C_r}. \tag{19}$$

The depth-wise temporal convolution mixes nearby time steps so that each input to the state update sees a small neighborhood, while the SiLU activation keeps the magnitude under control for the recurrent scan.

Bidirectional state update with learnable decay. The context is scanned forward and backward over time by a state-space recurrence. At each time step t, a learnable decay $a \in (0,1)^{C_r}$ controls how much of the previous state is retained. The decay is produced from a learnable parameter θ and a non-negative scalar ξ,

$$a = \sigma(\theta \cdot \xi), \qquad 1 - a = e^{-\xi}, \tag{20}$$

so that a and $1-a$ are coupled and both lie in $(0,1)$. The forward and backward states are updated by

$$h_t^f = a \odot h_{t-1}^f + (1-a) \odot W\tilde{u}_t, \qquad h_0^f = 0, \tag{21}$$

$$h_t^b = a \odot h_{t+1}^b + (1-a) \odot W\tilde{u}_t, \qquad h_{T+1}^b = 0, \tag{22}$$

where W is a learnable input mapping and $\tilde{u}_t$ is the context at time t. The forward state accumulates information causally from the past, and the backward state accumulates from the future; together they cover the entire temporal context. The learnable decay a gives the state a controllable memory horizon: when a is close to 1 the state retains most of its history (long memory), and when a is close to 0 the state is dominated by the current input (short memory).

Selective gate. The two states are concatenated along the channel dimension,

$$h_t = \big[\, h_t^f;\ h_t^b \,\big] \in \mathbb{R}^{B\times T\times 2C_r}, \tag{23}$$

and then element-wise multiplied by the gate stream G, giving a selectively gated content

$$h_t^{\mathrm{sel}} = h_t \odot G. \tag{24}$$

The multiplication by G is what makes the module selective rather than a plain linear recurrence: at each time step, the gate stream decides which channels of the processed state should contribute, so the model can emphasize time steps that look relevant and suppress ones that look noisy.

Temporal channel gate and reshape. The selectively gated content is passed through an MLP and a sigmoid to produce the temporal channel gate,

$$G_c = \sigma\big(\mathrm{MLP}(h_t^{\mathrm{sel}})\big) \in \mathbb{R}^{B\times T\times C_r}. \tag{25}$$

Finally, G_c is reshaped to the spatial layout of the lower-branch feature,

$$\tilde{G}_c = \mathrm{Reshape}(G_c) \in \mathbb{R}^{B\times C_r\times T\times 1\times 1}, \tag{26}$$

so that it can be broadcast-multiplied with the spatial feature L_d in Equation (9). The gate G_c reweights the locally refined motion feature: it amplifies channels whose long-term temporal response indicates the action and suppresses channels that respond only to static appearance, letting the aggregated state memory influence which motion features the final representation keeps.

Discussion. TSA combines three ingredients that are usually studied separately: a content stream, an input-dependent selection gate, and a long-term state memory with a learnable decay. The selective multiplication by G implements the input-dependent selection that distinguishes Mamba-style selective state-space models from plain linear recurrences, while the learnable decay a gives the state a controllable memory horizon. The whole module operates in the reduced bottleneck dimension C_r, so its cost is linear in the sequence length and its parameter footprint is small. The mechanism is related to recent state-space video models Park et al. (2024) in spirit, but here it produces only a temporal channel gate rather than a full feature, so it does not need the heavy selective scan machinery used in those models.

The two branches complement each other. MoDeA operates at the short time scale and on spatial positions: it returns a refined local feature that carries the fine motion of individual body parts. TSA operates at the long time scale and on channels: it returns a per-channel gate that carries how the action's overall response develops. Each branch is incomplete on its own. Short-term deformation without long-term context cannot tell whether a movement is a transient jitter or a sustained part of the action, and long-term memory without short-term refinement cannot distinguish actions whose overall response is similar but whose local motion differs. Their product combines both views: the gate G_c selects, channel by channel, which parts of the locally refined motion feature contribute to the final representation, which is then up-projected and added to the stage feature.

3.3 Complexity and training

The framework keeps the additional cost over a plain 2D backbone modest. The frozen CLIP encoder runs without gradients and is shared across all stages, so it contributes only forward-pass cost. The STG block operates in a reduced bottleneck dimension and confines attention to local windows, so its cost per stage is $O\big(B\cdot C_r\cdot T\cdot H\cdot W\cdot w^2\big)$, linear in the spatial resolution. The MoDeA branch samples a down-sampled lattice, so its attention cost is proportional to the number of sampled positions rather than the full resolution, and the TSA branch is linear in T. Because the two motion branches operate in the bottleneck and never materialize a full-resolution intermediate tensor, both parameters and memory stay low; we report concrete counts in the appendix.

The final stage feature is pooled spatially and temporally and passed to a linear classifier. We train end-to-end with a label-smoothed cross-entropy loss. Smoothing softens the target distribution and prevents over-confidence on classes that are visually close, a common failure mode in fine-grained recognition where two classes may share nearly identical poses. Formally, the target $\tilde{y}$ mixes the one-hot label y with a uniform distribution,

$$\tilde{y}_c = (1-\varepsilon)\, y_c + \frac{\varepsilon}{K}, \tag{27}$$

where K is the number of classes and ε the smoothing factor. The frozen encoder, the alignment modules, the STG blocks, and the DG-TDM blocks are trained jointly, with the frozen branch kept at zero gradient.

4 Experiments

We evaluate DG-TDMon two motion-centric action recognition benchmarks, Diving48 and Something-Something V2 (SSv2). Below we first describe the datasets, evaluation protocol, and implementation details, then compare DG-TDMwith existing methods on the two benchmarks, and finally present ablation studies on Diving48 that isolate the contribution of each branch.

4.1 SETUP

Datasets. We evaluate on two benchmarks. Diving48 Li et al. (2018) contains approximately 18k trimmed video clips of 48 dive classes; its videos share similar pool backgrounds and differ in subtle body movements such as rotation angle and timing. We use the official V2 train/test split. Something-Something V2 Goyal et al. (2017) is a large-scale motion-centric benchmark of around 220k videos in 174 classes of everyday object manipulations, such as pushing, pulling, and dropping; the classes are defined by the direction and manner of the motion rather than by the objects or the scene, so success requires temporal reasoning over the whole sequence. For both datasets we report top-1 and top-5 accuracy.

Implementation details. We use a ResNet-50 backbone as the temporal action encoder. The frozen spatial branch is a CLIP ViT-B/16 encoder whose parameters are kept frozen. The DG-TDM module is inserted after the 3×3 convolution of the last bottleneck in each of the four stages, with a reduction ratio of 4; the STG block is inserted after each of the four stages, with a window size of 7. We train with stochastic gradient descent, momentum 0.9, weight decay 5×10^{-4}, and a step schedule. The classifier uses label-smoothed cross-entropy with smoothing 0.1. Training is performed on two NVIDIA V100-SXM2-32GB GPUs. Our main setting samples $T = 8$ frames per video. Full hyperparameters are listed in the appendix.

Baselines. We compare against representative methods that report results on Diving48 and SSv2. Reported numbers of the compared methods are taken from their original papers (with the source cited in the table).

4.2 COMPARISON WITH STATE OF THE ART

Tables 1 and 2 compare DG-TDMwith existing methods on the two benchmarks. We report values exactly as published in the original papers, together with the input configuration (frames $\times$ resolution) used by each method. The compared methods span specialized fine-grained models (DSTS, CAN), lightweight 2D temporal models (TANet, TEINet, GST, AdaFuse, TDN, and AK-Net), and recent adaptation baselines (MANet, LoKi, and AIM). For Diving48, the numbers under the DSTS protocol are as reported by DSTS Li et al. (2022c) and those under the DVT protocol by DVT Wang and Torresani (2022); the remaining rows come from the original publications. For SSv2, all baseline values are taken from the cited original papers and are restricted to configurations with an explicitly reported 8-frame input. We evaluate DG-TDMon both benchmarks with 8 frames.

Under the $8 \times 1 \times 1$ single-clip protocol, DG-TDMreaches 83.55%/97.82% top-1/top-5 on Diving48 and 65.3%/89.1% on SSv2. Its top-1 accuracy is the highest among the compared methods with a ResNet-50 backbone on both benchmarks.

4.3 ABLATION STUDIES

We ablate DG-TDMon Diving48, the benchmark whose classes share the most similar backgrounds and therefore stresses localization the most. All ablations use 8 frames. We organize the analysis into three parts: the progressive build-up of the framework (Table 3), the internal design choices of each branch (Table 4), and the sensitivity to the main hyperparameters (Table 5).

Progressive build-up. Table 3 adds the components of DG-TDMto a TSN-style baseline (a plain ResNet-50 that samples frames, extracts 2D features, and classifies their temporal average), one component at a time, using the $8 \times 1 \times 1$ protocol of the main comparison. Each step isolates one mechanism: MoDeA contributes short-term cross-frame alignment, TSA adds a persistent long-term state and thereby completes DG-TDM, and the STG branch finally brings in the frozen-CLIP prior as guidance.

MoDeA raises top-1 by 7.73 points over appearance averaging alone, confirming that following the moving body part across frames is essential when the discriminative signal is a small local displacement. TSA adds a further 2.48 points, showing that a state that aggregates the whole sequence captures information that local alignment leaves out. The STG branch contributes the final 1.98 points, which measures the value of bilateral interaction and adaptive gating under the same temporal encoder. Top-5 accuracy improves consistently along the sequence, with the full model reaching 97.82%.

Table 1: Comparison on Diving48. Metrics are top-1 and top-5 accuracy (%). The Input / views column preserves the notation used by each source: an explicit $frames \times clips \times crops$ value is shown when reported, while sources that report only the number of input frames retain a single frame-count value. Resolution-based configurations are also kept in their original form. Top-5 values are marked "-" where the source does not report them. "†" marks the DSTS protocol Li et al. (2022c) and "‡" the DVT protocol Wang and Torresani (2022); the two papers use different evaluation protocols, so values across the two groups are not directly comparable. The TDN row reports the corrected official result compiled in Chen et al. (2022), using the updated Diving48 annotations. The CAN row takes the 8-frame ablation configuration from Li et al. (2025a), whose main result uses 16 frames; the AIM row reports the 8-frame result reproduced by Cai et al. (2025).

Method	Backbone	Input / views	Top-1 (%)	Top-5 (%)
Protocol of DSTS Li et al. (2022c)				
Video Swin-B Liu et al. (2022)	Swin-B	-	80.5†	-
Swin-B + DSTS Li et al. (2022c)	Swin-B	-	83.0†	-
Protocol of DVT Wang and Torresani (2022)				
DVT Wang and Torresani (2022)	ViT-B	16×224^2	83.5‡	-
Methods with lightweight backbones				
TDN Wang et al. (2021) (corrected)	ResNet-50	8	80.5	-
AK-Net (TSM) Chen et al. (2022)	ResNet-50	8	79.3	-
AK-Net (TDN) Chen et al. (2022)	ResNet-50	8	81.7	-
CAN Li et al. (2025a) (8-frame config)	ResNet-50	$8\times1\times1$	82.7	98.3
MANet Li et al. (2025b)	ResNet-50	$8\times1\times1$	81.8	98.0
LoKi Cai et al. (2025)	ViT-B/16	$8\times1\times1$	78.4	98.2
AIM Yang et al. (2023)	ViT-B/16	$8\times1\times1$	77.4	97.8
Ours				
DG-TDM	ResNet-50 + frozen CLIP	$8\times1\times1$	**83.55**	**97.82**

Table 2: Comparison on Something-Something V2 with 8 input frames and lightweight ResNet-50 backbones. Metrics are top-1 and top-5 accuracy (%). TANet follows its original $8\times1\times1$ protocol Liu et al. (2021); TDN and AK-Net follow the unified 8-frame comparison in AK-Net Table II Chen et al. (2022); ILA, STAN-conv, and M2-CLIP follow their reported multi-view 8-frame settings Tu et al. (2023); Liu et al. (2023); Wang et al. (2024b). FEASE-Net, CAN, and MANet use their reported $8\times1\times1$ configurations Zhou et al. (2024); Li et al. (2025a;b). A dash denotes a top-5 value not reported by the source.

Method	Backbone	Frames	Top-1 (%)	Top-5 (%)
ResNet-50 and lightweight baselines				
TANet Liu et al. (2021) (IN)	ResNet-50	$8\times1\times1$	60.5	86.2
TEINet Liu et al. (2020) (IN)	ResNet-50	8×1	61.3	-
GST Luo and Yuille (2019) (IN)	ResNet-50	8×1	61.6	-
AdaFuse Meng et al. (2021) (IN)	ResNet-50	8×1	59.8	-
TDN Wang et al. (2021) (IN, corrected)	ResNet-50	8×1	64.0	-
AK-Net (TSM) Chen et al. (2022) (IN)	ResNet-50	8×1	60.2	-
AK-Net (TDN) Chen et al. (2022) (IN)	ResNet-50	8×1	64.3	-
Recent 8-frame adaptation baselines				
ILA Tu et al. (2023) (ViT-B/16)	ViT-B/16	$8\times4\times3$	65.0	89.2
STAN-conv Liu et al. (2023) (CLIP)	ViT-B/16	$8\times1\times3$	65.2	90.5
M2-CLIP Wang et al. (2024b) (CLIP)	ViT-B/16	$8\times1\times3$	66.9	90.1
FEASE-Net Zhou et al. (2024)	ResNet-50	$8\times1\times1$	64.2	88.8
CAN Li et al. (2025a)	ResNet-50	$8\times1\times1$	63.9	89.1
MANet Li et al. (2025b)	ResNet-50	$8\times1\times1$	63.6	88.8
DG-TDM(ours)	ResNet-50 + frozen CLIP	$8\times1\times1$	**65.3**	**89.1**

Design choices. Table 4 switches off one internal design choice at a time from the full model, so that each row keeps the affected component present and isolates a single factor. The overall contribution of the MoDeA branch is already measured by Table 3 (removing it recovers the $+7.73$ points of the

Table 3: Progressive ablation on Diving48 with $8 \times 1 \times 1$ views. Each row adds one component of DG-TDMto the previous configuration; the last column reports the top-1 gain over the preceding row. Metrics are top-1 and top-5 accuracy (%).

Configuration	Top-1 (%)	Top-5 (%)	ΔTop-1
TSN-style baseline (ResNet-50)	71.36	95.82	—
+ MoDeA (short-term motion alignment)	79.09	96.74	+7.73
+ TSA (long-term state memory; DG-TDM complete)	81.57	97.77	+2.48
+ STG guidance (full DG-TDM)	**83.55**	**97.82**	+1.98

Table 4: Design-choice ablation on Diving48 with $8 \times 1 \times 1$ views. Each row switches off one internal design choice of the full DG-TDMwhile keeping the affected component and the remaining configuration fixed. Metrics are top-1 and top-5 accuracy (%).

Configuration	Top-1 (%)	Top-5 (%)
Full DG-TDM(reference)	**83.55**	**97.82**
DG-TDM: additive branch outputs (w/o TSA gate selection)	83.41	98.02
TSA: forward-only scan (w/o backward context)	82.58	97.76
STG: one-directional guidance (w/o $Y \to V$)	82.70	97.88
CLIP prior: last-stage-only guidance (w/o per-stage)	81.83	97.65

baseline row), so the rows below address the choices inside the remaining machinery. One factor is the fusion of the two DG-TDM branches, which are combined multiplicatively so that TSA selects the MoDeA-refined local features with a per-channel gate; we compare this with adding the two branch outputs in parallel. A second factor is the direction of the TSA scan, and we remove its backward pass. The bilateral nature of the STG guidance is a third factor, and we keep only the prior-to-feature direction. The last factor is where the frozen CLIP prior is applied: we test injecting it at the last stage only, asking whether the position knowledge must be used progressively at several spatial scales or once at the top. The full model serves as the reference row.

Hyperparameter sensitivity. Table 5 verifies that the reported gain of the guidance is robust to the two settings that define how the STG block injects the frozen CLIP prior into the recognition branch: the window size w and the residual bound $\lambda_{\max}$, swept around the defaults of the full model (w=7, $\lambda_{\max}$=0.2; last row of Table 3). The window size fixes the spatial extent of the bilateral attention: too small a window cannot cover the moving action region, while too large a window lets background context dilute the prior. The residual bound caps how strongly a stage may be modified: a low bound starves the injection, and a high bound lets the prior overwrite the appearance features of the lower branch. Because the residual scale is itself learned within $[0, \lambda_{\max}]$, the bound only sets the allowed headroom. Across both sweeps the default configuration stays on a plateau in top-1: the window w=7 outperforms both of its neighbors, and tightening the residual bound to 0.1 is the only change that costs a measurable amount, while loosening it to 0.4 brings no further gain, confirming that the guidance does not depend on a fine-tuned setting.

4.4 Efficiency analysis

Because the frozen CLIP branch does not back-propagate and the DG-TDM operates in a reduced bottleneck dimension, the framework adds modest trainable parameters and FLOPs over the temporal action encoder baseline. The appendix reports the parameter count, FLOPs, and per-clip throughput of the main variants.

4.5 Qualitative analysis

The learned mechanisms are inspected on sampled Diving48 frames in two ways. The deformation offsets predicted by MoDeA show whether different channel groups track different body-part motions, and the gates produced by the STG branch show whether the lower branch focuses on the action region instead of the shared background.

Table 5: Hyperparameter sensitivity of the STG guidance on Diving48 with 8 frames. The starred value marks the default of the full model (Table 3); each hyperparameter is swept to one smaller and one larger value. Metrics are top-1 and top-5 accuracy (%).

Hyperparameter	Value	Top-1 (%)	Top-5 (%)
STG window size w	5	83.29	97.74
	7*	**83.55**	**97.82**
	9	83.14	97.82
Residual bound $\lambda_{\max}$	0.1	82.95	97.38
	0.2*	**83.55**	**97.82**
	0.4	83.37	98.05

5 Conclusion

We presented DG-TDM, a three-branch framework for fine-grained action recognition. The task is hard because the classes look almost identical in appearance, and the difference lies in how the body moves. We decomposed this into two problems and gave each to a separate branch. A frozen CLIP encoder handles localization, supplying stable spatial priors learned from large-scale pre-training, and a spatio-temporal guidance branch uses them to focus a lightweight temporal encoder on the action region. Because a 2D backbone has no temporal modeling of its own, a dual-granularity temporal dynamics module adds it on the 2D feature stream: MoDeA tracks short-term body-part motion with dense deformable sampling, and TSA keeps a long-term memory over the sequence to characterize the overall temporal response. The framework thus targets the two difficulties that make fine-grained actions hard, namely similar backgrounds and subtle body movement, without inflating the backbone into a 3D network.

Ablations on Diving48 confirm that the two temporal branches and the guidance all carry part of the accuracy: every variant that removes the deformable sampling of MoDeA, turns the TSA state into a forward-only scan, or disables the per-stage prior injection drops below the full model. The frozen CLIP expert could in principle be replaced by a cheaper spatial encoder trained end-to-end; we leave that exchange, together with an extension to further benchmarks, to future work.

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